ADULT CENSUS INCOME PREDICTION

Random Forest Classification Project | Radha Krishan Tripathi | LNCT University | 08 November 2026

Problem statement
Predict whether an individual's annual income is **>50K** or **<=50K** using demographic and employment attributes. This is a binary classification task.

Dataset
Adult Census Income (Kaggle). The loaded file contained **32,561 rows and 15 columns**; target: **income**.

PREPROCESSING & MODEL WORKFLOW

Data inspection
32,561 records; 15 original columns. Six numerical and nine string/categorical columns, including the income target.

Missing values
The '?' entries were treated as missing in workclass (1,836), occupation (1,843), and native.country (583). Each was filled using that column's most frequent category.

Encoding & features
Separated X from y, then applied one-hot encoding with drop_first=True. The feature matrix became **32,561 × 97**.

Train-test split
80:20 split with random_state=42: training **26,048** rows; testing **6,513** rows. Trained a RandomForestClassifier with 100 trees.

MODEL PERFORMANCE

85.11%	71.72%	60.90%	65.87%
Accuracy	Precision*	Recall*	F1-score*

*Precision, recall and F1-score are for the positive class **>50K**. Confusion matrix (actual rows × predicted columns): TN = 4,607; FP = 369; FN = 601; TP = 936. The classification report showed F1 = 0.90 for <=50K and 0.66 for >50K.

KEY FINDINGS

1. The model correctly classified 5,543 of 6,513 test records (85.11% accuracy); 970 predictions were incorrect.

2. Performance was stronger for the <=50K class (precision 0.88, recall 0.93, F1 0.90) than for >50K (0.72, 0.61, 0.66).

3. The model's largest feature-importance values were fnlwgt (0.1652), age (0.1581), capital.gain (0.0906), hours.per.week (0.0855), and marital.status_Married-civ-spouse (0.0824).

4. Of the 970 errors, 601 were actual >50K records predicted as <=50K; 369 were actual <=50K records predicted as >50K.

CONCLUSION

A Random Forest classifier was built, evaluated, and saved using Joblib. It achieved 85.11% accuracy, but recall for the >50K class (60.90%) indicates that higher-income cases are harder for this model to identify. Future work could compare class-weighted training, tune hyperparameters, and use cross-validation before selecting a final model.

DATA SOURCE

Kaggle – Adult Census Income dataset (uciml/adult-census-income). Dataset file used: **adult.csv**.